

1. Introduction

Objective:

This report presents an in-depth analysis of the Shaking Bed Air Supply System. The goal is to identify operational inefficiencies, potential risks, and optimization opportunities to enhance process performance, minimize energy consumption, and maintain product quality.

Context:

The Air Supply System is responsible for regulating key parameters such as air temperature, fan speeds, and steam valve operations. Maintaining these parameters within optimal ranges ensures efficiency and prevents system failures.

2. Data Analysis

Key Findings:

- **Temperature Fluctuations:** Observed high variations ($>10^{\circ}\text{C}$) at fan speeds below 25%.
- **Optimal Stability:** Temperature fluctuations are lowest at fan speeds between 50-60%.
- **Energy Inefficiency:** Higher fluctuations at low fan speeds lead to unnecessary steam valve modulation.
- **Data Anomalies:** Some extreme values identified and corrected to ensure accuracy.

Correlation Analysis:

- A strong negative correlation between **fan speed and temperature fluctuation**.
- Higher fan speeds correlate with **reduced energy consumption and stable air temperature**.
- Steam valve openings show an inverse relationship with fan speed, suggesting potential inefficiencies at lower speeds.

3. Data Visualization

Key Visualizations:

- **Scatter Plots:** Illustrating temperature fluctuation vs. fan speed.
- **Heatmaps:** Showing correlations among fan speed, air temperature, and steam valve operation.
- **Box Plots:** Highlighting variations in system performance at different fan speed levels.
- **Bar Charts:** Comparing temperature stability across fan speed ranges.

4. Optimization Opportunities

Recommendations:

1. Fan Speed Adjustment Strategy

- Maintain fan speed **between 50-60%** for optimal temperature stability.
- Avoid operating below **25% fan speed** due to excessive fluctuations.

2. Predictive Maintenance Implementation

- Install **real-time sensors** to monitor fan speed, temperature, and steam valve operations.
- Develop a **machine-learning-based predictive maintenance model** to anticipate failures before they occur.

3. Energy Efficiency Improvement

- Reduce reliance on **steam valve modulations** by maintaining **optimal air circulation through fan speed adjustments**.
- Implement **automated control logic** that dynamically adjusts fan speed based on real-time temperature variations.

5. Deliverables

Documents & Reports:

- **This Final Report** summarizing the analysis, findings, and recommendations.
- **Detailed Data Analysis Report** including code snippets and methodology.

Visuals & Data Files:

- **Graphs & Charts** (scatter plots, heatmaps, bar charts, etc.)

Code & Execution Files:

- **CSV/Excel data files** used for analysis.

6. Conclusion

The Shaking Bed Air Supply System exhibits inefficiencies primarily at lower fan speeds. Maintaining fan speed within 50-60% significantly enhances system stability and reduces energy waste. Implementing a predictive maintenance approach will further optimize operations, prevent failures, and improve overall performance.